1. Clinical Question & Core Hypothesis

Clinical Question

CEACAM5 is upregulated in tumors and associated with higher immune cell infiltration and MSI-H, yet increased TILs do not always translate into better patient outcomes. Why does CEACAM5-associated immune infiltration sometimes fail to confer a survival benefit? 2) Tier 1 paths used: (a) Tumors → CEACAM5 (CEACAM5 upregulation in tumors) and (b) CEACAM5 → Higher immune cell infiltration (association with increased immune infiltrate). These form the mechanistic backbone linking a tumor-expressed molecule to an immune-rich microenvironment. 3) Tier 2 support: the MSI-H association with higher immune infiltration supports the idea that CEACAM5-high tumors may overlap with immunogenic (MSI-H) biology; gene expression as the measurement modality supports testing transcriptional correlates of immune function. 4) Endpoint choice: Overall survival (OS) was selected because it is the most robust, widely available clinical endpoint in TCGA and best for testing whether immune infiltration associated with CEACAM5 yields a meaningful clinical benefit. 5) Deliberate exclusions: I excluded the imaging modality 'ultrasound' and the organ context 'bladder' because they are peripheral to the central tumor-immune mechanism; I also excluded specific immune-exhaustion or MMR genes (e.g., PDCD1, MLH1) to keep the hypothesis focused on the CEACAM5–TIL axis, placing MSI-H into secondary_modifiers rather than the core claim?

Working Hypothesis

CEACAM5 upregulation in tumor cells is associated with increased tumor-infiltrating lymphocyte (TIL) density, an effect amplified in MSI-H tumors. However, CEACAM5-expressing tumors drive a qualitatively impaired TIL phenotype (e.g., higher exhaustion/regulatory markers and lower cytotoxic function), such that high TIL density does not translate into improved overall survival (OS). Specifically, CEACAM5-high/TIL-high/MSI-H tumors will show OS similar to or worse than CEACAM5-low/TIL-high/MSI-H tumors, and will exhibit higher expression of TIL exhaustion markers (e.g., PD-1, LAG-3, CTLA-4) and lower cytotoxic effector signals (e.g., GZMB, PRF1) within the TIL compartment. This interaction between CEACAM5 expression, TIL functional state, and MSI-H status will modulate survival and, in contexts with available therapy-response proxies, immunotherapy responses, such that CEACAM5-high tumors show attenuated benefit from immune checkpoint inhibitors despite high TIL density.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	CEACAM5
Primary endpoint	OS

2. Why This Gene — Not Others

Primary hypothesis gene(s)

CEACAM5: CEACAM5-associated immune infiltration sometimes fail to confer a survival benefit? 2) Tier 1 paths used: (a) Tumors → CEACAM5 (CEACAM5 upregulation in tumors) and (b) CEACAM5 → Higher immune cell infiltration (association with increased immune infiltrate).

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	MSI-H	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: Retrospective analysis in TCGA-like cohorts (RNA-seq for CEACAM5; immune-infiltration scores from deconvolution; MSI-H status; OS; and, when available, ICI-response proxies). Stratify by CEACAM5 (high/low) and by MSI-H status; evaluate the interaction effect of CEACAM5 and TIL density on OS using Cox models, adjusting for confounders (age, stage, subtype).

Prediction 2: Immunophenotyping readouts: examine expression of TIL functional markers (exhaustion: PDCD1, LAG3, CTLA4; cytotoxic: GZMB, PRF

Prediction 3: and composition of T cell subsets via deconvolution to test the mechanistic link between high TIL density and dysfunctional TILs in CEACAM5-high tumors.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.
- Multi-modal validation pipeline: computational -> spatial -> functional, providing mechanistic depth beyond association.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- MSI-H: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.